

A Positively Generated Function System with No Uniform Complete Positive-Generation Bound

Run fa_banach_001 — lane 2

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Status: candidate counterexample, likely valid, human review required. Question 7.3 of Humeniuk–Kennedy–Manor asks whether positive generation of a function system forces complete α -generation for one uniform α . The explicit Cantor-cube function system below is positively generated but has no such uniform constant, giving a negative answer.

1 Source question

Humeniuk, Kennedy, and Manor study dualizability and positive decomposition for possibly nonunital operator systems in arXiv:2312.04791, later published in the *Journal of Functional Analysis* **289** (2025), 111153 [1]. Their published Question 7.3, on arXiv PDF page 24, is:

[15, Theorem 14.3], and so ρ cannot be completely bounded below.

By [15, Theorem 3.9], the map ρ is completely positive. But, by definition of ρ , the map ρ is a complete order isomorphism if and only if every positive map $S \rightarrow M_n$ is completely positive, but this is not true even for finite dimensional operator systems S .

So, even if S is a function system that is a dualizable operator system, its dual S^* is typically not *completely* order and norm isomorphic to a function system, and never can be when S is infinite dimensional. We don't know whether positive generation of a function system S is enough to guarantee completely bounded positive generation and so dualizability. We leave this as an open question.

Question 7.3. If $S \subseteq C(X)$ is a positively generated function system, is S completely α -generated for some $\alpha > 0$?

If so, then S is dualizable if and only if it is positively generated. In Proposition 8.4 below, we show that positive generation actually guarantees positive generation at all matrix levels. If Question 7.3 has a negative answer, then by Proposition 8.4 below there is a function system S for which each $M_n(S)$ is α_n -generated, but the sequence (α_n) cannot be chosen to be bounded.

In Example 8.6 below, we give a matrix ordered operator space which is positively generated, but not completely α -generated for any $\alpha > 0$. We do not know a function system with this property.

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Here a function system may be nonunital. An ordered $*$ -Banach space E is α -generated if every $x \in E^{\text{sa}}$ has a decomposition $x = y - z$ with $y, z \in E^+$ and

$$\|y\| + \|z\| \leq \alpha \|x\|.$$

A matrix ordered operator space is completely α -generated if the same constant works in every $M_n(E)$.

2 Proof Intuition

The source already constructs a matrix ordered version of $\text{MIN}(\ell^1)$ whose positive cones are coordinatewise and whose complete positive-generation constants are unbounded. Its unresolved issue is that the example is not known there to be a function system.

The key observation is that an equivalent norm makes precisely this ordered space concrete and commutative. On the Cantor cube $K = \{0, 1\}^{\mathbb{N}}$, send $z \in \ell^1$ to the continuous subset-sum function $f_z(\varepsilon) = \sum_k \varepsilon_k z_k$. Singleton points recover each coefficient, so positivity at every matrix level is exactly coordinatewise positivity. The resulting norm is equivalent to $\|z\|_1$, and both the concrete function system and $\text{MIN}(\ell^1)$ are minimal operator spaces. Thus their coefficient identification is a completely bounded complete order isomorphism. A uniform generation constant would transfer through it, contradicting the source's example.

3 Counterexample

Let

$$K = \{0, 1\}^{\mathbb{N}}$$

with the product topology. For $z = (z_k) \in \ell^1$ define

$$f_z(\varepsilon) = \sum_{k=1}^{\infty} \varepsilon_k z_k, \quad \varepsilon = (\varepsilon_k) \in K,$$

and put

$$S = \{f_z : z \in \ell^1\} \subseteq C(K).$$

Theorem 1. *The space S is a closed, selfadjoint, positively generated function system. It is not completely α -generated for any $\alpha > 0$. Consequently, Question 7.3 has a negative answer.*

We separate the one external obstruction used in the proof.

Lemma 2 (Humeniuk–Kennedy–Manor). *Let $X = \text{MIN}(\ell^1)$, equipped at every matrix level with the coordinatewise positive cone*

$$M_n(X)^+ = \{(A_k)_{k \geq 1} : A_k \in M_n^+\}.$$

Then X is not completely β -generated for any finite β .

Proof. This is Example 8.6 on arXiv PDF pages 28–29 of the source [1]. For completeness, its obstruction is recalled here. The dual matrix ordered operator space is $X^* = \text{MAX}(\ell^\infty)$ with the usual coordinatewise ordering. If X were completely β -generated, the source's Proposition 5.5 would make X^* completely 2β -normal.

The minimal and maximal operator-space structures on the infinite-dimensional space ℓ^∞ are not completely equivalent [2, Theorem 14.3]. Hence, for arbitrarily large R , there are n and $u \in M_n(\ell^\infty)$ with

$$\|u\|_{\min} \leq 1, \quad \|u\|_{\max} > 4R.$$

In the C^* -algebra $M_n(\ell^\infty)$, write u as the linear combination of the four positive and negative parts of $\text{Re } u$ and $\text{Im } u$. Each of these four positive elements has minimal norm at most one, and one of them, say y , has maximal norm greater than R . Thus

$$0 \leq y \leq 1, \quad \|1\|_{\max} = 1, \quad \|y\|_{\max} > R.$$

This violates every uniform complete normality constant. Taking $R > 2\beta$ gives the desired contradiction. $\square$

Proof of the theorem. Step 1: S is a closed function system. The series defining f_z converges uniformly, because its tail is bounded by the corresponding ℓ^1 tail. Thus $f_z \in C(K)$. The map $z \mapsto f_z$ is complex linear, injective (evaluate at the singleton points), and respects conjugation.

We have the upper estimate

$$\|f_z\|_\infty \leq \|z\|_1.$$

For the reverse estimate, given $\theta \in [0, 2\pi]$, let

$$E_\theta = \{k : \operatorname{Re}(e^{-i\theta} z_k) \geq 0\}$$

and let $1_{E_\theta} \in K$ be its characteristic sequence. Then

$$|f_z(1_{E_\theta})| \geq \sum_k \max\{\operatorname{Re}(e^{-i\theta} z_k), 0\}.$$

The average over θ of the k th summand is $|z_k|/\pi$. Tonelli's theorem therefore gives some θ for which

$$\|f_z\|_\infty \geq \frac{1}{\pi} \|z\|_1.$$

The coefficient map is consequently a Banach-space isomorphism, so its image S is closed. Hence S is a concrete (possibly nonunital) function system.

Step 2: determine all matrix cones. If $f_z = f_z^*$, singleton evaluations show that every z_k is real. Furthermore,

$$f_z \geq 0 \text{ on } K \iff z_k \geq 0 \text{ for every } k.$$

The forward implication again uses singleton points, and the reverse implication is immediate. Thus $f_z = f_{z^+} - f_{z^-}$ for selfadjoint f_z , and S is positively generated.

More generally, an element $F \in M_n(S)$ has a unique absolutely summable coefficient sequence (A_k) in M_n and

$$F(\varepsilon) = \sum_k \varepsilon_k A_k.$$

If $F \geq 0$, evaluation at the singleton point $e_k \in K$ gives $A_k \geq 0$. Conversely, if every $A_k \geq 0$, every subset sum above is positive. Hence

$$M_n(S)^+ \text{ is exactly the coordinatewise cone for every } n.$$

Step 3: compare completely with the obstruction space. Let $X = \operatorname{MIN}(\ell^1)$ with the coordinatewise matrix cones from the lemma, and define the coefficient map

$$T : S \longrightarrow X, \quad T(f_z) = z.$$

Step 2 says that T and T^{-1} preserve positivity at every matrix level, so T is a complete order isomorphism.

Every concrete subspace of a commutative C^* -algebra has its minimal operator-space structure [2, Proposition 3.3.1]; hence S is minimal, while X is minimal by definition. The universal property of minimal operator spaces says that every bounded map into a minimal operator space is completely bounded with the same norm [3]. The scalar estimates from Step 1 therefore imply

$$\|T\|_{\text{cb}} \leq \pi, \quad \|T^{-1}\|_{\text{cb}} \leq 1.$$

Step 4: transfer a hypothetical uniform constant. Suppose that S were completely α -generated. Given $x \in M_n(X)^{\text{sa}}$, put $s = T_n^{-1}x$ and write $s = p - q$ with $p, q \geq 0$ and

$$\|p\| + \|q\| \leq \alpha \|s\|.$$

Then $x = T_n p - T_n q$ is a positive decomposition in $M_n(X)$ and

$$\|T_n p\| + \|T_n q\| \leq \|T\|_{\text{cb}} \alpha \|T^{-1}\|_{\text{cb}} \|x\| \leq \pi \alpha \|x\|.$$

The same constant works for every n , so X would be completely $\pi\alpha$ -generated. This contradicts the lemma. Therefore no finite complete generation constant exists for S . $\square$

4 Verification

The reusable script `code/verifier.py` exhaustively evaluates all subsets in seeded finite-coordinate examples. It checks the exact real norm formula, the complex $1/\pi$ lower bound, and the matrix singleton-cone mechanism. Run it with

```
conda run --no-capture-output -n sandbox python \
  runs/fa_banach_001/solutions/counterexamples/\
  2312.04791_cantor_cube_positive_generation_no_uniform_complete_bound/\
  code/verifier.py
```

These finite checks are sanity tests, not a proof of the infinite-dimensional MIN/MAX obstruction. The complete proof is the argument above. The separate `VERIFIER_REPORT.md` records the review checklist and output.

5 Novelty, limitations, and review recommendation

A bounded search through 11 August 2026 checked the run’s four lightweight indexes, the exact wording of published Question 7.3, the phrases “positively generated function system”, “completely alpha-generated”, and “MIN(ℓ^1) function system”, and later sources citing arXiv:2312.04791. The 2025 JFA version still states the question as open. No later answer or matching Cantor-cube construction was found. Novelty is therefore plausible but not certified.

The counterexample is nonunital, exactly within the source’s convention that function systems may be nonunital. It does not address a strengthened unital variant. It also does not quantify the optimal generation constant at each fixed matrix level.

Human review is recommended, with priority on two standard but decisive points: the complete-order identification of the coordinate cones, and the minimal-operator-space universal property used to promote scalar norm equivalence to cb norm equivalence.

References

- [1] A. Humeniuk, M. Kennedy, and N. Manor, *An extension property for noncommutative convex sets and duality for operator systems*, J. Funct. Anal. **289** (2025), 111153; arXiv:2312.04791.
- [2] V. Paulsen, *Completely Bounded Maps and Operator Algebras*, Cambridge Studies in Advanced Mathematics 78, Cambridge University Press, 2002.
- [3] D. P. Blecher, *Introduction to Operator Spaces*, lecture notes, Fields Institute; minimal-operator-space universal property in Section 1.2.